Non-LLM Routing: Accuracy–Cost Trade-offs

Jonathan-aligned outputs · HotpotQA: 4,814 train / 1,110 validation / 1,481 test · Shared standard lenient

Qwen2.5-7B / 72B retrieval + Llama-3.2-3B / Llama-3.1-70B answering · Three-seed probability ensembles

SS BASELINE **76.77% accuracy** 100.00% cost = 929 Table 5 budget units Direct strong retrieval + strong QA

Linear Softmax

TF-IDF + statistics → 2-way linear head

	Accuracy-first		SS-budget*	
Training	Acc. %	Cost %	Acc. %	Cost %
SFT	**76.50**	107.78	**69.75**	49.52
SFT + Reward	**76.64**	107.94	**71.37**	60.32
GRPO	**76.03**	106.60	**74.41**	96.59

Accuracy-first SS: SFT 1,463 · Reward 1,469 · GRPO 1,447

MLP · 64 hidden units

TF-IDF + statistics → 64 ReLU → 2

	Accuracy-first		SS-budget*	
Training	Acc. %	Cost %	Acc. %	Cost %
SFT	**76.84**	106.76	**75.89**	94.35
SFT + Reward	**76.30**	106.87	**76.03**	98.26
GRPO	**76.70**	108.21	**75.83**	95.54

Accuracy-first SS: SFT 1,450 · Reward 1,423 · GRPO 1,475

MLP · 128 → 64

TF-IDF + statistics → 128 → 64 → 2

	Accuracy-first		SS-budget*	
Training	Acc. %	Cost %	Acc. %	Cost %
SFT	**76.84**	107.46	**76.77**	98.18
SFT + Reward	**76.84**	106.75	**76.77**	98.33
GRPO	**76.77**	108.61	**69.55**	46.84

Accuracy-first SS: SFT 1,461 · Reward 1,448 · GRPO 1,481

Accuracy-first: highest validation accuracy, cost tie-break. SS-budget*: highest validation accuracy with validation cost ≤ 100% SS.

WHY CAN COST EXCEED 100%?

Direct SS: 814 + 115 = 929 (100%)

Weak-first SS: 77 + 814 + 3 + 115 = 1009 (108.61%)

Accuracy-first uses SS on 96–100% of questions and also pays for weak calls. Budgeted MLP-128→64 SFT matches SS accuracy at 98.18% cost.

01 SFT

Learn W/S labels with weighted cross-entropy.

Retrieval: coverage in weak selected evidence.

QA: keep correct weak answers; upgrade if rescued.

Both-wrong cases map to W.

3 epochs · lr 1e−4 · checkpoint by val AUC

02 SFT + Reward

Start from SFT; score both cached W/S actions.

R = token F1 − 0.1 × C_train / 929

Maximize exact expected R + 0.03 × entropy.

Refine both stages; RET uses best downstream QA R.

3 epochs · lr 1e−5 · checkpoint by val reward

03 GRPO

Start from SFT; sample 8 actions per row.

Group-normalized advantages + policy gradient.

Add 0.02 KL to frozen SFT reference; no PPO clip.

Stage adaptation of Jonathan’s public code.

2 epochs · lr 1e−4 · checkpoint by val reward

* Budget constraints apply on validation; test cost is observed, not guaranteed. Both operating points use the same trained model. Amber costs exceed SS.

Cost excludes decision-model compute. Reward prices follow the author: [80,195,894,929]; actual weak-first evaluation: [80,195,894,1009]. GRPO is an adaptation, not a verified slide reproduction.

Previously inspected test: descriptive results. 76.84% is only one additional correct answer vs SS. Train-fitted features only; no LLM encoder or generation.